



2.4 Transformations of graphs

Functions · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The graph of $y=af(b(x-c))+d$ is obtained from $y=f(x)$ by: a vertical stretch by factor a (reflection in x -axis if $a<0$), a horizontal stretch by factor $1/b$ (reflection in y -axis if $b<0$), a horizontal translation by c , and a vertical translation by d . Translations (c,d) are applied to x and y individually, while stretches scale distances from the relevant axis.

Worked example: follow the reasoning

Describe the transformation of $y=x^2$ to $y=2(x-3)^2+1$, and state the new vertex.
Starting vertex $(0,0)$: horizontal translation of 3 (right) gives vertex $(3,0)$; vertical stretch factor 2 (from x -axis) keeps vertex at $(3,0)$; vertical translation of 1 (up) gives final vertex $(3,1)$.
New vertex: $(3,1)$.

Quick check · multiple choice

The graph of $y=f(x-2)+3$ is obtained from $y=f(x)$ by:

- A Translation 2 right, 3 up
- B Translation 2 left, 3 down
- C Translation 2 right, 3 down
- D Translation 2 left, 3 up

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 The graph of $y=f(x)$ has a minimum point at $(2,-1)$ and passes through $(0,3)$. [7]
- (a) State the coordinates of the minimum point of $y=f(x-1)+4$.
 - (b) State the coordinates of the minimum point of $y=-2f(x)$.
 - (c) Given $f(x)=x^2-4x+3$, verify the original minimum point $(2,-1)$, then write the equation of the transformed graph $y=f(x-1)+4$ in the form ax^2+bx+c .

Final answer: _____



Answers and explanation

Question 1 [7]

(a) Translation right 1, up 4: minimum becomes $(2+1, -1+4) = (3,3)$.

(b) Vertical stretch factor 2 and reflection in x-axis: y-coordinate of minimum becomes $-2(-1)=2$, and since reflection turns a minimum into a maximum, the point $(2,2)$ is now a maximum.

(c) $f(x)=x^2-4x+3=(x-2)^2-1$, vertex $(2,-1)$, confirmed. $f(x-1)+4 = ((x-1)-2)^2-1+4 = (x-3)^2+3 = x^2-6x+9+3 = x^2-6x+12$.

Quick check answer

A. Replacing x with $(x-2)$ translates right by 2; adding 3 translates up by 3.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.